YIFEI SUN

Principal Software Engineer · Backend & Platform Architecture · 13 Years

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SUMMARY

Staff-level engineer with 13 years of experience and a track record of rapidly building and shipping 0→1 products from concept to production. Founding backend engineer of CarGurus' innovation lab (2020–present), where I designed and launched multiple new products under high ambiguity — including an MCP server for ChatGPT integration, digital commerce platforms, and GraphQL federation. Combines deep technical expertise across the full stack with strong product intuition, thriving in fast-paced environments where ideas move from prototype to real users in days. Experienced influencing technical roadmaps across 10+ teams and 3 departments.

LEADERSHIP HIGHLIGHTS

- Founding backend engineer of innovation lab — built and shipped multiple 0→1 products from concept to production
- Led cross-org initiatives spanning 10+ teams; translated ambiguous requirements into shipped products
- Rapidly prototyped and delivered MCP/ChatGPT integration as a high-visibility proof-of-concept in weeks
- Mentored senior engineers; defined architectural patterns and standards adopted across organizations

Technologies: Java, Spring Boot, GraphQL (Netflix DGS), Python, Node.js · AWS, Kubernetes, Kafka, Redis, PostgreSQL · Rapid Prototyping, Full-Stack Product Development, Microservices, Event-Driven, MCP/LLM Integration · Datadog, Grafana · CI/CD, Docker

INDEPENDENT PROJECT

GenAI Decision Engine & Experimentation Platform

- Architected a hybrid rules + multi-path GenAI engine (fast vs. deep inference) with a policy overlay over heterogeneous telemetry, producing explainable real-time recommendations.
- Designed a configurable policy framework with DB-backed enterprise configs, environment-specific overrides, and a scheduler-driven tuning loop that sweeps parameters and publishes promotable config versions.
- Built "Prompt Lab" for historical backfill, end-to-end decision simulation, offline evaluation/backtesting, and prompt/parameter search (grid & Bayesian optimization) with a robust scheduler, observability, and multi-provider LLM support.

EXPERIENCE

CarGurus — Principal Software Engineer / Founding Backend Engineer, Innovation Lab Oct 2019 – Present

ChatGPT Apps SDK MCP Server (0→1 — Concept to Production in Weeks)

- Sole backend engineer who championed, designed, and implemented an MCP server compatible with the OpenAI Apps SDK, exposing CarGurus capabilities into ChatGPT as a high-visibility proof-of-concept.
- Went from initial concept to working demo in days; iterated rapidly on ambiguous requirements with PMs and directors; designed for safe, incremental production rollout alongside existing APIs and clients.
- Served as the key engineering representative for AI product strategy, engaging with senior leaders across the company to shape the technical roadmap for LLM integration.

Digital Shopping & Online Leads Products (0→1 — Multiple Product Launches)

- As founding backend engineer of the innovation lab, designed and launched a new digital shopping product during the pandemic — from blank repo to production in weeks, capturing online leads via microservice and event-driven architecture.
- Subsequently built and shipped multiple product experiments in rapid succession, enabling the company to quickly test, launch, and safely retire product bets without destabilizing core systems.
- Thrived in high-ambiguity environment; balanced speed of delivery with architectural quality, ensuring experimental products could be promoted to core platform when successful.

GraphQL Federation — Delivery Lookup DGS Subgraph

- Designed and integrated a Netflix DGS subgraph into CarGurus' federated GraphQL gateway, exposing vehicle delivery data to frontend, backend, and AI consumers from a single graph — supporting the new AI chatbot/shopping assistant platform.
- Implemented schema federation, batch data loaders (N+1 prevention), and JVM warmup strategies; built load tests and staging integration tests.

High-Traffic Legacy API Modernization

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core API behind CarGurus' most-visited page (~8,000 rpm); produced initial architecture in ~2 weeks, shipped first milestone in 6 months.
- Consolidated 12 downstream integrations into 5 well-defined services via a new orchestration layer and GraphQL boundary.

p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)

Cross-Org Platform & API Leadership

- Established patterns for API design, caching, and distributed orchestration adopted across multiple teams; authored key design docs, RFCs, and decision records that drove consensus across platform and product teams.
- Mentored multiple senior engineers on system design, performance work, and path to Principal-level scope.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Authentication & Identity Modernization

- Upgraded legacy authentication to a modern Keycloak-based SSO platform, unifying auth across internal systems, an enterprise partner platform, and a large external community. Introduced social login and mobile login; migrated multiple Angular, Node.js, and Java monoliths behind the new auth gateway with minimal downtime.

Event Sourcing & CQRS Initiative

- Drove early design and prototyping of a microservice architecture based on event sourcing and CQRS; defined foundational patterns for services, events, and read/write models adopted by new domains.

Cross-Org Collaboration

- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations with clear observability and rollback paths.

Toast Inc. — Senior Software Engineer

Sep 2014 – Oct 2017

Monolith → Microservices Decomposition

- Among the earliest engineers during hypergrowth (first customer to hundreds in a few years); designed and implemented core backend services (Order Service, Labor Service, foundational APIs) that became the basis for future growth.
- Built initial AWS deployment pipelines before a dedicated DevOps team existed; created the first automated testing framework before dedicated QA, enabling safer releases during rapid product expansion.

Third-Party Integrations & Cross-Functional Leadership

- Led integrations with key partners (e.g., workforce management), ensuring reliable bidirectional sync and schema evolution. Partnered with PMs, sales, and support to translate early customer feedback into platform capabilities.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010